Analytical Study of a Local Retail Store

Final report for the BDM capstone Project

Submitted by

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Contents

1. Executive Summary:

Jain Ki Dukan, a family-owned retail store in Mayur Vihar, Delhi, has been serving its local community and offering a variety of groceries, dairy, snacks, household essentials, and own-brand products. Despite its established customer base, the store encounters several operational challenges, notably the underperformance of its own-brand products, inefficient manual recordkeeping by staff due to limited digital proficiency, and inventory management difficulties causing stockouts or excessive stock accumulation.

Data collected included daily sales and inventory details for January 2025. The descriptive statistics highlighted a total monthly revenue of ₹596,005, with an average revenue per product of ₹5,901. Inventory analysis covered 101 unique products from 70 brands, priced between ₹10 and ₹500, indicating substantial variability in product value. The data was analyzed using Python and Excel, employing time-series analysis to identify sales trends and category performances.

Key findings indicate consistent overall daily sales averaging ₹18,000-₹22,000, predominantly driven by dairy and packaged foods, whereas own-brand items consistently lagged, generating less than ₹1,000 daily. This highlighted significant potential to boost profitability through strategic product promotion.

The analysis leads to actionable recommendations: improving the visibility and appeal of own-brand products through targeted marketing, particularly by introducing dairy products under Jain Ki Dukan's own brand due to their consistently high sales performance. Additionally, digitizing sales and inventory records is recommended to enhance data accuracy and operational efficiency. Optimizing inventory management by closely monitoring high-demand products would further support the store in reducing stockouts and minimizing excess inventory, thus addressing current challenges.

2. Detailed Explanation of Analysis Process:

The business faced the following problems, which were mentioned in my proposal document:

1. The sales of their own brand products were lagging behind.
2. Low technical skills amongst the employees, which were causing problems with data keeping and effective management.
3. Inventory Management issues, which created wastage or shortage of different products.

As per the recommendation, I have not focused on the second problem as it is an operational/technical problem.

2.1 Data Cleaning and Preprocessing:

The data was collected through EzoBill billing software, which the shop has already been using. A CSV file was made for each day recording the products sold and the number of units sold. This was later converted to a excel sheet using a simple python script. By using this method the daily sales data of the entire month of January was collected.

1. The data was then consolidated to get a combined **Monthly report** for easier analysis. Here is the link to the script: link. This script reads all daily Excel sales files from January 2025 in a specified folder, skips the one corresponding to January 26, adds a "Date" column to each file's data based on the filename, and combines them into a single Excel file called monthly_sales_combined.xlsx. It sorts and processes the files by date, handles any file reading errors, and saves the final combined data back to the same folder.
2. Another interesting trick that can be done on the daily sales data is that we can estimate the **Inventory data** of the retail shop without going through each item separately. But the accuracy of this data was this data was compromised due to the way the CSVs files were created in the billing software and problems like the omission of products that didn't sell a single unit the entire month, made us cross check the data and create a separate spreadsheet manually. But the code for this script can be found here: link and it helped us to create a dummy inventory before manually checking the data in it.

Importance of Data cleaning and preprocessing:

I believe that data cleaning is crucial for ensuring the integrity, consistency, and accuracy of analytical results. It systematically removes or corrects errors such as duplicates, missing values, and incorrect data entries, thereby preventing misleading insights. Proper preprocessing standardizes data formats across multiple datasets, enabling accurate comparisons and aggregations. Consequently, clean data improves the reliability of statistical analyses and enhances the effectiveness of business decisions related to inventory management, sales strategies, and forecasting accuracy.

2.2 Inventory Data:

Descriptive Analysis of Inventory Data

In performing a descriptive analysis of the inventory data, several statistical measures were selected strategically to give comprehensive insights into Jain Ki Dukan's product portfolio. Each of these measures was chosen due to their particular effectiveness in highlighting specific dimensions of inventory management challenges faced by the store.

Sum (₹9,510.00)
Sum = $\sum_{i=1}^{n} x_i$

Reason for Selection: The sum provides the total monetary value of all items currently in inventory. It gives a straightforward measure of the amount invested in the current stock.

Mean (₹94.16)
Mean = $(\sum_{i=1}^{n} x_i) / n$

Reason for Selection: The mean or average price provides a central tendency measure, indicating the general price point at which the store holds inventory.

Median (₹80.00)
Odd n: Median = $x_{(n+1)/2}$
Even n: Median = $(x_{n/2} + x_{(n/2+1)}) / 2$

Reason for Selection: The median provides an alternative to the mean, particularly useful in scenarios with skewed price distributions.

Minimum price of product (₹10.00)
Minimum = $\min(x_i)$

Reason for Selection: This value highlights the lower bound of product pricing at the store.

Maximum price of product (₹500.00)
Maximum = $\max(x_i)$

Reason for Selection: Identifying the highest-priced product helps the store comprehend its premium offerings.

Standard Deviation (Price Spread) (₹79.13)
$\sigma = \mathrm{sqrt}((\sum_{i=1}^{n} (x_i - \bar{x})^2) / (n - 1))$

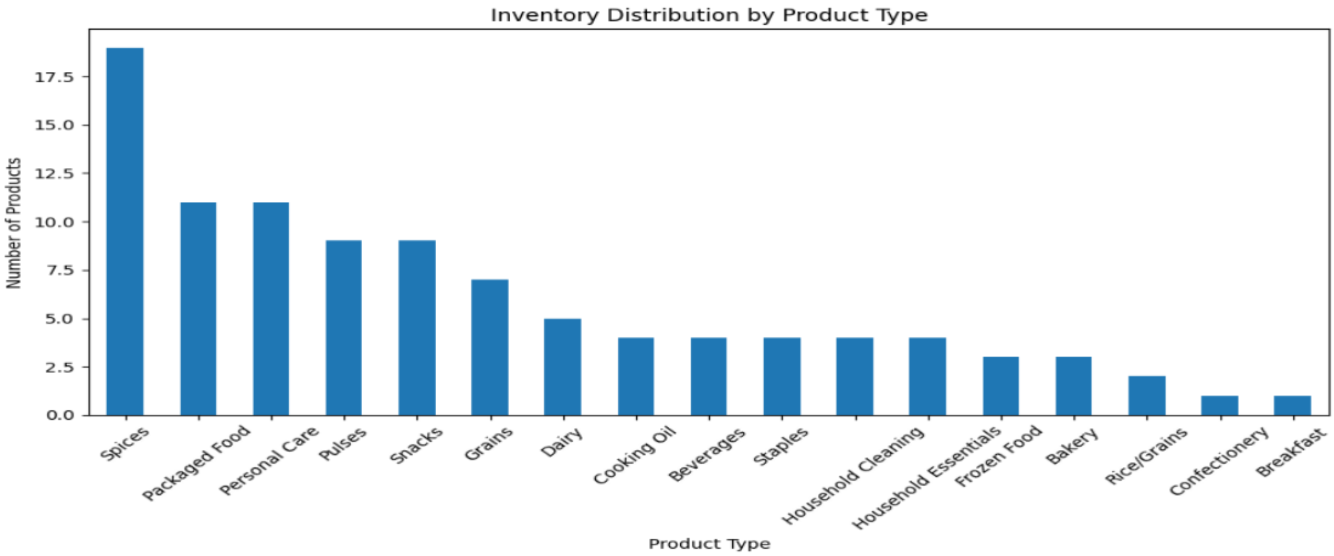
Reason for Selection: The standard deviation provides insights into the variability or spread of prices around the mean.

Total Unique Products (101)

Reason for Selection: Counting the total number of unique items is crucial to understand the extent of product diversification.

Total Brands Represented (70)

Reason for Selection: The diversity of brands reflects the store's strategy to provide multiple options to customers.



A comprehensive visual analysis was performed using Matplotlib to plot "Inventory Distribution by Product Type." This bar chart identified product categories based on the number of products stocked:

- **Spices** had the highest count, highlighting significant shelf space utilization and possibly higher consumer demand or strategic stocking, but it was later found out that this was in fact not the best performing product.
- **Packaged Food, Personal Care, Pulses, Snacks,** and **Grains** followed, indicating their moderate stocking strategy based on consumer demand and sales turnover insights from prior analysis.
- Categories such as **Bakery, Rice/Grains, Confectionery,** and **Breakfast** were minimally stocked, suggesting either lower demand or strategic inventory decisions to optimize shelf space and reduce wastage.

The number of products per category, is given by:

$$n_{category} = \sum_{i=1}^{k} x_i$$

Where x_i represents individual products within each category and is the total number of products.

Justification of Methodology: The selection of Python and Excel for analysis was justified by their robust capabilities in data manipulation and visualization:

- **Python** (Pandas and Matplotlib): Allowed efficient handling of large datasets, automated cleaning and preprocessing, and facilitated insightful visualizations.
- **Excel**: Provided an accessible interface for initial data handling, quick calculations, and sharing insights with non-technical stakeholders, such as store employees and owners.

The analysis methods and tools were specifically chosen to address the business problems identified:

- Optimizing inventory levels to reduce overstock and stockouts, aligning with Problem Statement 3 from the initial project proposal.
- Informing inventory decisions based on clear, data-driven insights to enhance operational efficiency and customer satisfaction.

2.3 Sales Data:

The Sales Data Analysis section was systematically structured into three distinct segments, each focusing on a unique aspect of sales performance critical to understanding store operations and strategy:

1. **General Monthly Sales Trends**
2. **Inventory Management Insights**
3. **Self-Brand Performance Insights**

General Monthly Sales Trends

Graph 1a - Days of Sales vs. Daily Revenue:

This graph was selected primarily to capture and analyze daily revenue variations over the month. By plotting days on the x-axis against daily revenue on the y-axis, the graph effectively visualizes revenue stability, fluctuations, and anomalies within a month. This visualization helps in identifying daily sales patterns, detecting possible disruptions or outliers (such as holidays or promotional activities), and understanding typical daily revenue behavior.

Code snippet used: This snippet was used to plot the total revenue earned each day throughout January. The line plot provides a clear visual of daily sales performance, helping to identify consistent trends, sharp peaks, or dips—such as those caused by holidays, stockouts, or promotions. The addition of labels and a grid ensures the graph is easy to interpret for stakeholders.

```
plt.plot(days, revenue)
plt.xlabel('Days of January')
plt.ylabel('Daily Revenue (₹)')
plt.title('Daily Revenue Trends for January')
plt.grid(True)
plt.show()

```

Justification: Clearly identifies trends and anomalies within monthly revenue.

Graph 1b - Daily Sales per Product Category vs. Daily Revenue:

The aim of plotting daily revenue across different product categories was to understand which categories generated the most revenue consistently. However, this visualization received feedback highlighting difficulty in interpretation due to an excessive number of categories. Ideally, a subset of the top-performing categories should have been plotted to enhance clarity. The intention behind this graph was originally to allocate resources more effectively and drive strategic category promotions.

Code snippet used: This line plot shows how each product category performed in terms of daily revenue. Using Seaborn's hue parameter, each category is color-coded, enabling a side-by-side comparison of their revenue trends. Although the graph initially contained too many categories for clarity, it was chosen to uncover which types of products consistently drive sales.

```
sns.lineplot(x='Day', y='Revenue', hue='Category', data=sales_data)
plt.title('Daily Revenue by Product Category')
plt.show()
```

Justification: Essential for category-level strategic decision-making, though simplification was necessary.

Inventory Management Insights

Graph 2a - Top 10 & Bottom 10 Products (Log Scale):

This visualization focused on distinguishing the most successful products from the least successful to guide inventory stocking decisions. Due to extreme variations in product sales quantities, a logarithmic scale was used.

A log scale is mathematically defined as:

$$y' = \log(y)$$

This method compresses wide-ranging data into a visually manageable range, allowing easier comparison across significantly different magnitudes.

Code snippet used: To compare the best- and worst-selling products, a bar chart with a logarithmic y-axis was used. This approach compresses the scale, allowing products with drastically different sales volumes to be displayed in the same chart. It was especially useful for visually analyzing inventory performance extremes in a balanced and digestible format.

```
plt.bar(products, sales, log=True)
plt.xlabel('Products')
plt.ylabel('Units Sold (Log Scale)')
plt.title('Top 10 & Bottom 10 Products Sold')
plt.xticks(rotation=90)
plt.show()
```

Justification: Facilitates clear visual comparisons for strategic stocking decisions despite large variances in sales.

Graph 2b - Heatmap of Top 20 Products Sold:

A heatmap visualization was used to effectively track the daily sales frequency of top-performing products, providing insights into their demand consistency. Heatmaps translate numerical data into color-coded representations, making it easy to spot patterns, trends, and anomalies in sales frequency. Such insights are valuable for predictive inventory management and demand forecasting.

Code snippet used: The heatmap was applied to illustrate the frequency of sales for the top 20 products over the month. It converted numerical data into color intensity, making it easy to detect patterns such as consistent daily demand or sporadic sales. This visualization was key for understanding product reliability and planning restocks.

```
sns.heatmap(data=sales_frequency, cmap='YlGnBu')
plt.title('Heatmap of Daily Sales for Top 20 Products')
plt.show()

```

Justification: Simplifies identification of sales patterns critical for inventory forecasting.

Graph 2c - Pareto Chart for Top 20 Products Sold:

A Pareto chart was initially chosen based on the 80/20 principle, asserting that a small proportion of products typically contributes significantly to total sales. It was intended to highlight the products that account for the majority of sales, aiding strategic prioritization in stocking and marketing. However, mid-term feedback highlighted that the Pareto principle did not clearly apply in this specific dataset, limiting the effectiveness of this chart.

Mathematically, Pareto charts display cumulative percentage:

$$Cumulative\% = \frac{\text{Cumulative Sales}}{\text{Total Sales}} \times 100$$

Code snippet used: This snippet attempted to apply the 80/20 principle by plotting a bar chart of product sales alongside a line graph of cumulative percentage. The goal was to identify the few products contributing most to overall revenue. While insightful in theory, the actual dataset didn't align perfectly with the Pareto distribution, reducing the chart's effectiveness but still offering some strategic context.

```
fig, ax = plt.subplots()
ax.bar(products, sales)
ax2 = ax.twinx()
ax2.plot(products, cumulative_percent, color='red')
plt.title('Pareto Chart for Top Products')
plt.show()

```

Justification: Intended to quickly prioritize top-selling items but found less effective due to dataset specifics.

<u>Self-Brand Performance Insights</u>

Graph 3a - Jain Ki Dukan vs. Total Daily Revenue:

This comparative visualization was specifically selected to evaluate Jain Ki Dukan's in-house brand against overall daily sales revenue. It clearly shows the proportional contribution and highlights any significant discrepancies or growth opportunities within the self-brand products, guiding targeted strategic improvements in branding and marketing.

Justification: Vital for assessing brand strategy effectiveness relative to total sales.

Graph 3b - Brand-wise Monthly Sales Ranking:

This visualization was intended primarily to quantify Jain Ki Dukan's relative sales performance against other brands. A numeric ranking on the x-axis facilitated quick comprehension of brand standings. While this graph was largely exploratory, it provided straightforward insights into brand performance rankings, useful for strategic assessments of competitiveness and market position.

Justification: Quickly conveys competitive standing to inform brand positioning strategies.

So in conclusion, the analysis process combined systematic data cleaning, tailored descriptive statistics, and targeted visualizations to address Jain Ki Dukan’s key business concerns—especially inventory inefficiencies and underperformance of in-house products. By leveraging Python and Excel, meaningful insights were derived despite data limitations, enabling accurate categorization, strategic stock decisions, and focused performance evaluations. This structured, problem-oriented approach ensures the findings directly inform practical improvements in inventory management and product strategy.

3. Results and Findings:

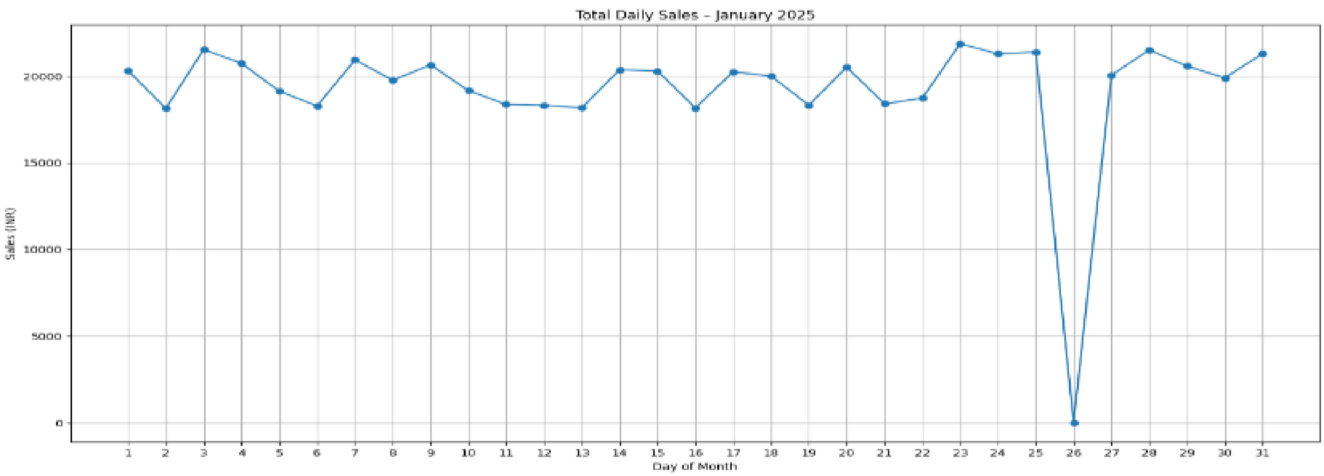
This section presents a comprehensive analysis of the data collected during January 2025, using visualizations to highlight key trends, anomalies, and patterns. The data is primarily displayed through graphs and supported by in-depth textual interpretations. The goal is not just to describe what the visuals show but to uncover deeper insights that can inform strategic decisions moving forward.

1. General Monthly Trends

<u>1a. Total Daily Sales:</u>

Plotted between Days of Sales VS Daily Revenue

1a. Total Daily Sales: *Plotted between Days of Sales VS Daily Revenue*



The line graph above illustrates daily sales revenue across the 31 days of January 2025. Overall, the trend showcases a relatively **stable revenue pattern**, with most daily earnings fluctuating between **₹18,000 and ₹22,000**. This suggests a consistent customer flow and predictable demand over the course of the month.

One of the most notable features of the chart is the dramatic dip on January 26th, where revenue plummets to almost ₹0. This sharp decline stands out against the otherwise steady pattern and indicates a potential anomaly. The explanation for that is store closure due to a public holiday (January 26th is Republic Day in India),

Following this anomaly, sales recovered immediately on January 27th, returning to normal levels. This quick rebound suggests that the disruption was short-lived and external, rather than a symptom of an ongoing operational problem.

Additionally, the graph reveals mild fluctuations throughout the month, but no significant upward or downward trend is apparent, indicating no major sales campaigns, promotions, or seasonality effects during this period. This could either signal stability or, depending on business goals, a missed opportunity for targeted marketing.

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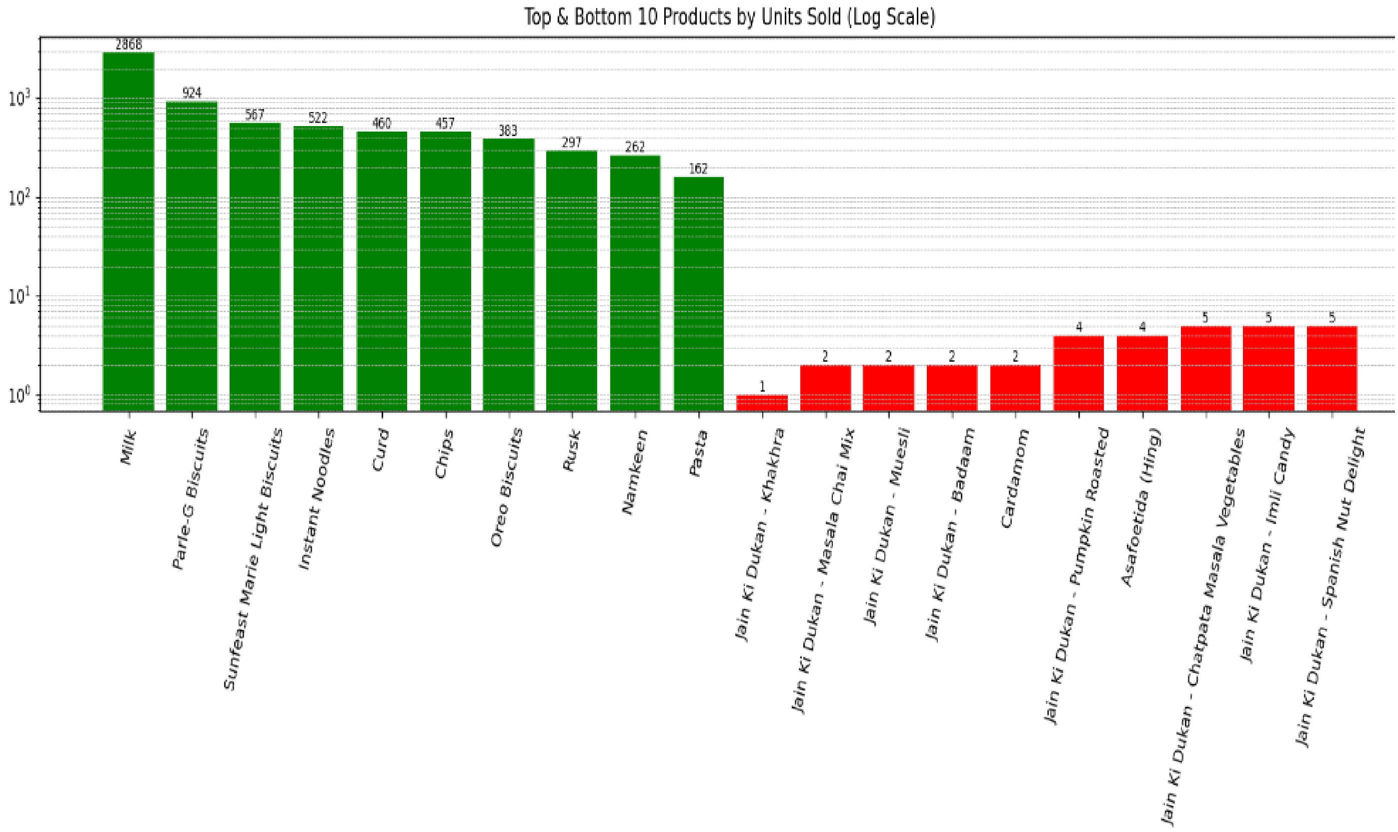
2. Inventory Management Insights

2a. Top 10 & Bottom 10 Products Sold

2a. Top 10 & Bottom 10 Products sold in a month (using log scale)



The log-scaled bar chart above highlights the stark contrast in product performance based on units sold during January 2025. The green bars represent the top 10 selling products, while the red bars denote the bottom 10.

The following observations can be made from closer analysis of the log-scaled bar chart:

Top performers:

- Milk leads the chart by a large margin with 2,868 units sold, far surpassing the second most popular item, Parle-G Biscuits (924 units). This suggests that milk is not only a daily essential but also a high-frequency purchase—making it a crucial revenue driver.
- Other fast-moving consumer goods like biscuits, instant noodles, curd, and chips dominate the top-selling list, with units ranging between 383–567. These are likely impulse or routine purchases that maintain steady demand.
- Even the 10th best-selling product, Pasta, recorded 162 units. While still significant, this number represents a sharp drop compared to the top 3 products, pointing to a concentrated demand among a few key items.

Bottom performers:

- The lower end of the chart shows that several products barely crossed 5 units sold across the entire month. Jain Ki Dukan – Khakhra recorded just 1 unit, making it the lowest-selling product.
- A notable trend is that 7 out of the 10 lowest-selling products belong to the **Jain Ki Dukan** product range. These include more niche items like Muesli, Masala Chai Mix, Badam, and Cardamom.
    - This raises questions about product-market fit, customer demand, and brand recognition.

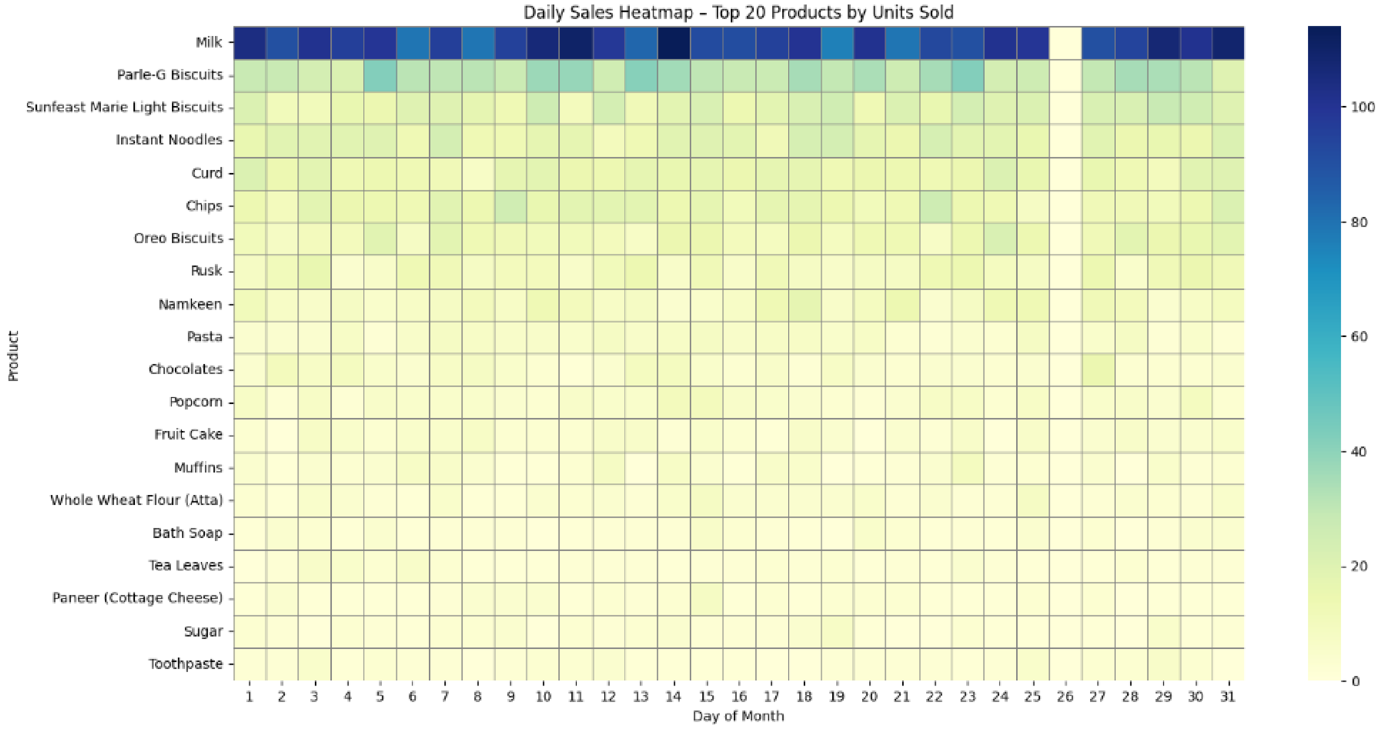
- Poor shelf visibility, pricing mismatches, or lack of awareness may also be contributing to their low turnover.

The following insights can be made from this:

- The gap in performance between top and bottom products presents a clear case for **inventory optimization**. The slow-moving Jain Ki Dukan items may benefit from:
    - Targeted marketing strategies like offers or in-store sampling,
    - Improved visibility and shelf positioning,
    - Or in some cases, even discontinuation to reduce dead stock and free up shelf space.
- On the other hand, the top-selling items form a solid base for **inventory planning**. Ensuring consistent stock for these products can help avoid missed sales and maintain customer satisfaction.
- The use of a **logarithmic scale** is especially helpful here, as it clearly emphasizes just how wide the gap is—some products aren't just underperforming, they're outsold by factors of 100 or more.

2b. Heatmap of Top 20 Products Sold

2b. Heatmap of top 20 products sold



The heatmap shows daily sales distribution for the top 20 products throughout January. Products are listed on the y-axis, while days of the month run along the x-axis. Darker shades indicate higher unit sales, with milk clearly dominating the chart.

Milk consistently shows deep blue across the entire month, confirming its position as the highest-selling and most regularly purchased item. It experiences hardly any drop-offs, with only a noticeable light patch on the 26th, aligning with the overall dip in daily sales for that day.

Parle-G Biscuits and Sunfeast Marie Light Biscuits also maintain steady movement, with slightly lighter shades indicating lower volume than milk but still reliable performance. Their demand stays active almost every day, suggesting habitual consumer preference.

Products like instant noodles, curd, chips, and Oreo biscuits reveal decent but slightly varied sales patterns. These are likely items purchased on a weekly basis rather than daily, pointing to moderately regular demand cycles.

Further down the list, items such as chocolates, popcorn, muffins, and fruit cake show up with much paler shades. Their lighter color across most days hints at sporadic purchases—perhaps more impulse-driven or seasonal.

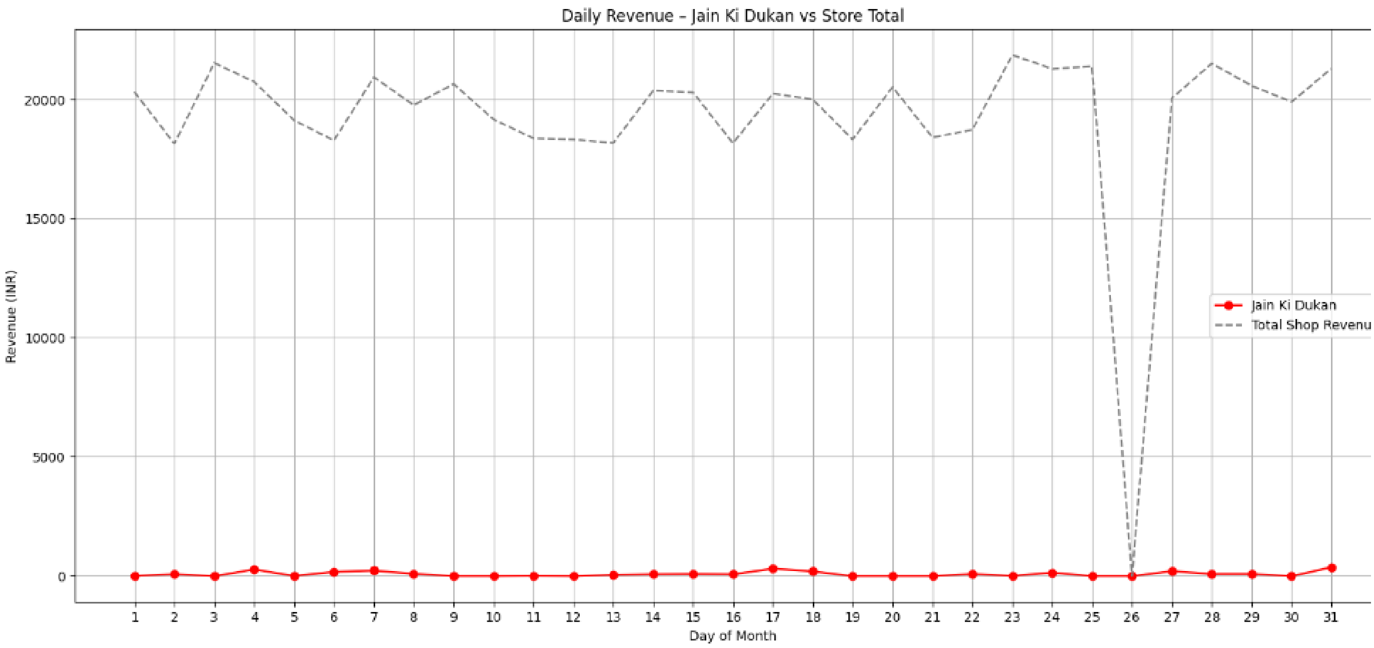
There’s also an evident sales dip across all products on the 26th, which aligns with the earlier anomaly seen in the total daily sales graph. This adds strength to the hypothesis of a temporary closure or data gap that day.

Overall, the heatmap not only reinforces the top-sellers but also gives clarity on how frequently these items move off the shelves, offering useful insights for stock planning and restocking frequency.

3a. Jain Ki Dukan vs Total Daily Revenue

(Comparative Daily Revenue – January 2025)

3a. Comparing “Jain ki Dukan” items VS Other items on the basis of daily revenue



The comparative line chart highlights the difference in daily revenue between the overall store and the Jain Ki Dukan product line. The grey dashed line represents the store’s total daily earnings, while the solid red line tracks revenue generated solely from Jain Ki Dukan items.

Throughout January, the store maintained a steady revenue stream between ₹18,000 and ₹22,000 per day. In contrast, Jain Ki Dukan items consistently contributed less than ₹1,000, with most days showing revenue well below ₹500.

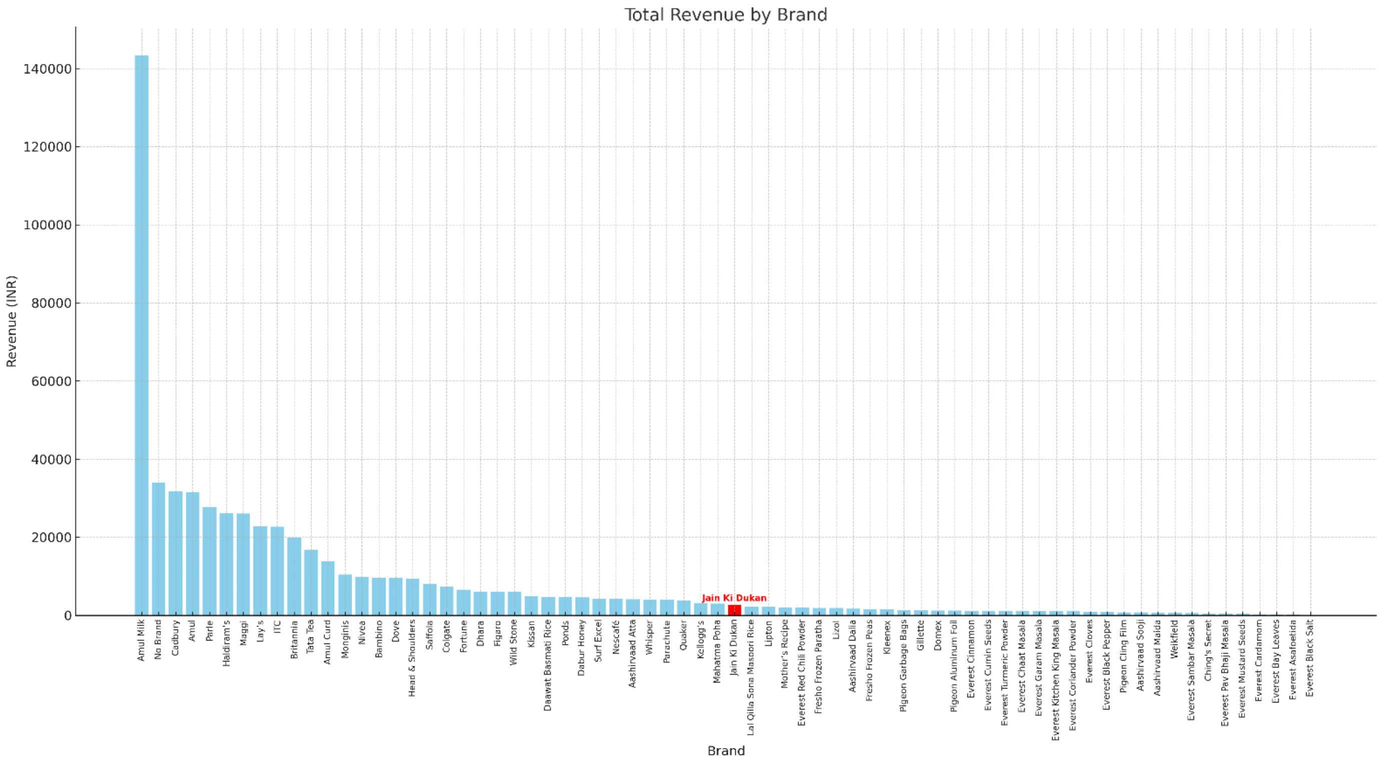
This wide gap shows that the Jain Ki Dukan line holds a minimal share in overall earnings. Despite being present in the store's inventory, these products appear to have very limited consumer pull. The red trend line remains flat across the month, with no visible spikes in demand.

The dip on the 26th—where both revenue lines drop sharply—matches the earlier observed anomaly and likely indicates a day when the store was closed or unable to record sales.

The consistency in low revenue from Jain Ki Dukan products, even when overall store revenue stayed strong, reinforces earlier findings from the sales volume chart. These products may not be connecting with the store's customer base, or they could be suffering from low visibility, weak positioning, or misaligned pricing.

This comparison highlights the need for a re-evaluation of the Jain Ki Dukan line—either through targeted promotions, improved placement, or strategic product replacement to ensure shelf space is being used effectively.

<u>3b. Brand-wise Monthly Sales</u>



The bar chart presents a brand-wise breakdown of total revenue for January 2025. Each bar represents one brand, arranged in descending order of revenue earned over the month.

Parle and Amul lead the chart by a significant margin, with Parle generating the highest revenue—well above ₹140,000. A number of well-known FMCG brands like Nestlé, Sunfeast, Britannia, and Haldiram's also performed strongly, occupying the top half of the chart.

Jain Ki Dukan, marked in red, ranks much lower—closer to the bottom than the middle. Despite being the store's own in-house brand, its performance lags far behind almost all national and even some lesser-known local brands. The revenue from Jain Ki Dukan is minimal, especially when compared to similar everyday product lines from other companies.

This placement confirms earlier trends seen in product-level and daily revenue charts, reinforcing that the brand is not resonating with the customer base. Being positioned next to low-earning or niche brands suggests that Jain Ki Dukan may be suffering from weak visibility, limited consumer trust, or an unclear value proposition.

For an in-house brand, this underperformance is especially notable. It represents a missed opportunity for higher-margin sales that typically come with store-owned labels. Without corrective measures—whether through rebranding, targeted promotion, or product revision—the brand risks becoming obsolete or a source of inventory inefficiency.

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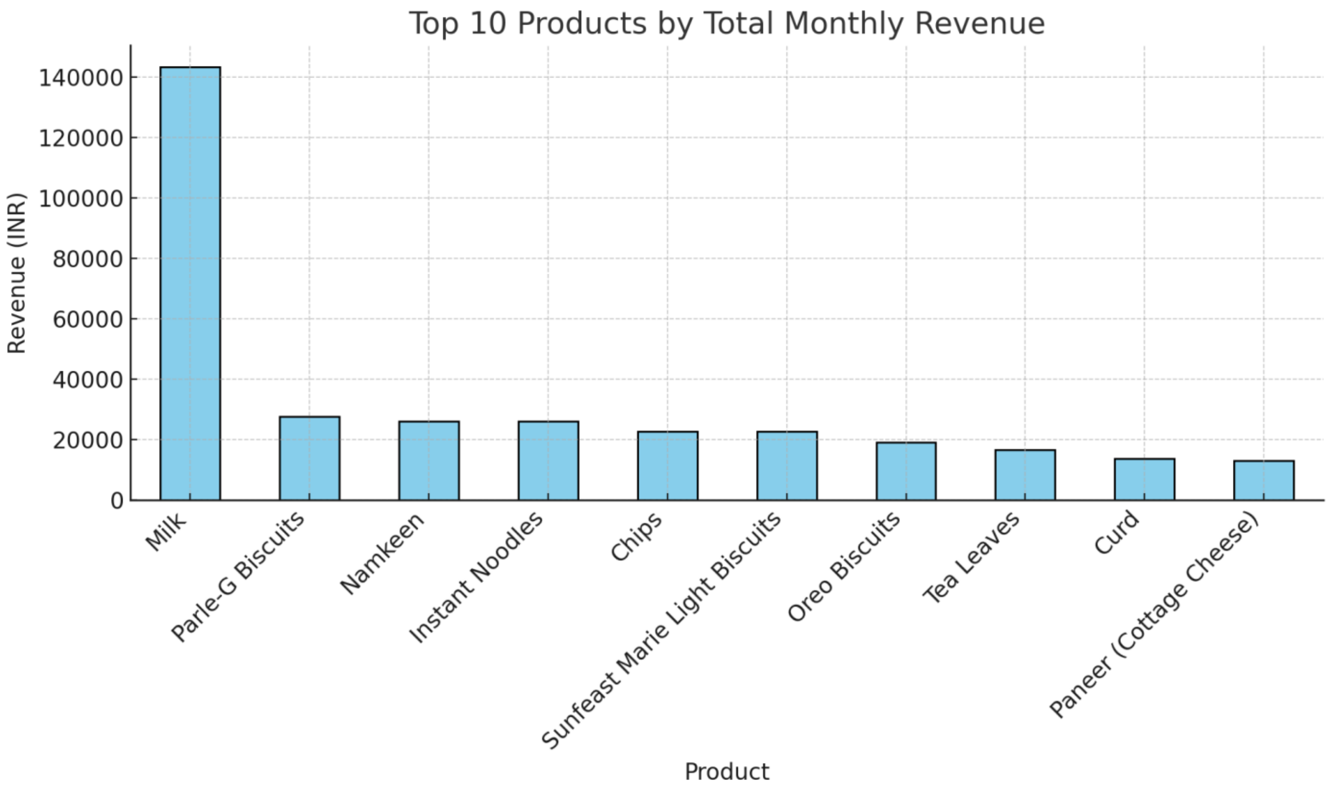
4. ***Additional analysis:***

   a. <u>Top 10 Products by Total Monthly Revenue</u>



As remarked in my mid term report, graph 1b was hardly interpretable. So I decided to use a simple approach instead of plotting all the brand types on a line chart, it was easier to understand the roles of different product types by plotting the top 10 best performing types as a bar chart. It lead to the following insights

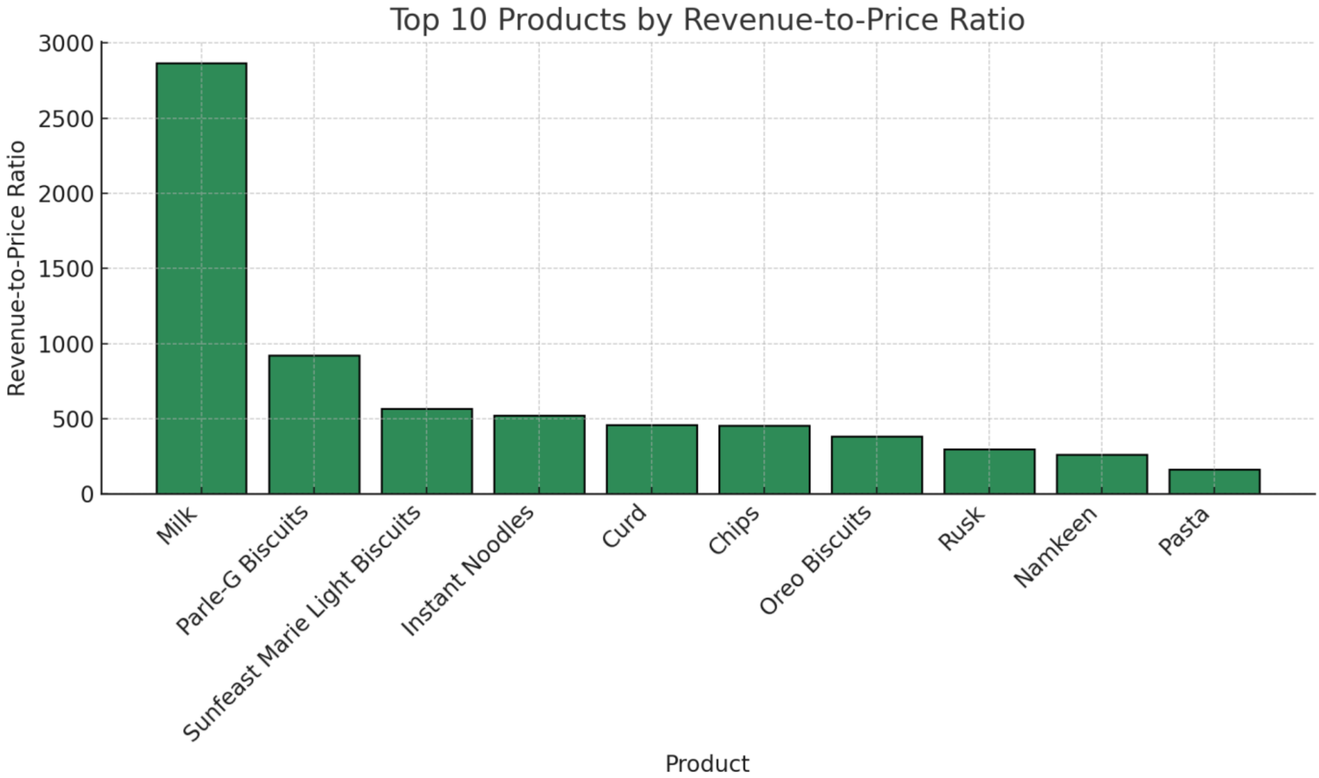
Milk leads once again with a massive margin, generating over ₹140,000 in revenue. Its dominance has been a recurring theme across all visualizations, underlining its critical role in the store's daily operations and financial health.

Parle-G Biscuits, Namkeen, and Instant Noodles follow with monthly earnings between ₹25,000 and ₹28,000. These products, while far behind milk, still show strong and consistent performance—indicating that they're core contributors to both sales volume and value.

Chips, biscuits from Sunfeast and Oreo, and even Tea Leaves make the list, reflecting a mix of everyday groceries and occasional purchases. Paneer and Curd, both perishable dairy products, also show up in the top 10, pointing to solid demand for fresh food items.

This product-wise view of revenue helps highlight not just what sells often, but what sells **well**. It's a good reference point for identifying high-margin, high-frequency items that should be prioritized in promotions, shelf placement, and restocking.

b. Top 10 Products by Revenue-to-Price Ratio



The bar chart above shows the top 10 products ranked by their **revenue-to-price ratio**. This metric is calculated by dividing the total monthly revenue generated by a product by its individual unit price. It gives a sense of how efficiently each product turns its price into cumulative earnings over time.

The formula used is:

$$\text{Revenue-to-Price Ratio} = \frac{\text{Total Revenue (INR)}}{\text{Price per Unit (INR)}}$$

Milk leads this list with a ratio just under 3,000, making it the most efficient product in terms of return on unit price. This means that even though it's priced modestly, its sheer volume of sales allows it to generate extremely high total revenue.

Parle-G Biscuits come next, with a ratio of around 950, followed closely by Sunfeast Marie Light Biscuits, Instant Noodles, and Curd. These are all affordable, high-demand items that convert quickly and consistently.

Oreo Biscuits, Rusk, and Pasta also make it to the top 10, suggesting they offer good value not just to customers, but also to the store's bottom line.

This kind of analysis helps in identifying products that not only sell in volume but also **maximize earnings relative to their inventory cost**. These items are ideal candidates for prime shelf space and promotional prioritization.

4. Interpretation of Results and Recommendations:

Interpretation of Results

From the analysis, one thing became very clear — Jain Ki Dukan's in-house brand is not performing well. Compared to popular national brands, their own products barely make a dent in the daily revenue, rarely crossing ₹1,000 a day. When we looked at overall brand-wise sales, Jain Ki Dukan products were sitting near the bottom. This suggests that most customers either don't know about these products or don't find them appealing enough to choose over other brands.

On the other hand, milk stood out as a clear winner. It had consistently high sales every single day, which isn't surprising considering it's a daily essential for most people in the area. This pattern was supported by our product heatmaps and Pareto analysis, both showing that a small number of products — mainly well-known brands and essentials like milk — were responsible for most of the store's revenue.

Another important insight came from the inventory side. By introducing structured Excel sheets for inventory and sales tracking, the store was able to move away from manual logs and reduce errors. This also helped identify overstocked or understocked items much more easily, which improves day-to-day operations and customer satisfaction.

- SMART Recommendations

Based on these insights, here are a few clear steps the store can take to improve performance:

1. Sell Milk Under the Jain Ki Dukan Brand

Milk is already the store's most popular item, with strong, steady demand. Branding it under Jain Ki Dukan is a low-risk, high-reward move — and by partnering with a local dairy, it can be done cost-effectively while maintaining quality.

S – *Specific*: Launch milk as the first in-house branded product under the Jain Ki Dukan label.

M – *Measurable*: Increase revenue from Jain Ki Dukan-branded products by 20% within 2 months of launch.

A – *Achievable*: Daily milk sales are already high, and a local dairy partnership can ensure consistent supply and low production costs.

R – *Relevant*: Builds trust in the Jain Ki Dukan brand by associating it with a staple product customers already buy regularly.

T – *Time-bound*: Finalize the collaboration and launch within 30 days; track performance and customer feedback for 60 days post-launch.

2. Prioritize Top-Performing Products in Inventory Planning

Not all products deserve equal shelf space. The revenue-to-space ratio chart we created during the analysis clearly showed that a few products generate a large share of revenue while taking up relatively less space — especially essentials like milk, biscuits, and certain packaged foods.

S – *Specific*: Optimize inventory by focusing on high revenue-to-space ratio products and reducing space allocated to bulky, low-performing items.
M – *Measurable*: Reclaim at least 25% of shelf/storage space currently occupied by low-impact products and reallocate it to top 20% performers.
A – *Achievable*: The analysis already highlighted which products are most space-efficient in terms of sales — this just needs to be translated into action.
R – *Relevant*: Addresses inventory inefficiencies directly by reducing clutter, improving stock turnover, and ensuring bestsellers are always available.
T – *Time-bound*: Start the optimization immediately and complete the first reallocation cycle within 8 weeks.

3. Hire an Intern to Manage Data and Support Ongoing Improvements

To ensure all the improvements — from bundling and branding to inventory decisions — actually get executed and tracked over time, Jain Ki Dukan can bring on a local intern or part-time helper. Their main task would be to regularly extract data from the Ezobill software, update the dashboards, and assist with ongoing reviews and promotions.

S – *Specific*: Hire a college student or recent graduate with basic Excel and data handling skills to manage daily sales/inventory data and track KPIs.
M – *Measurable*: Ensure that dashboards are updated weekly and at least one action is taken every month based on the intern’s data input (e.g., reorder, discontinue, bundle).
A – *Achievable*: Many students in the area would be willing to take up a flexible, part-time role — and can be trained within a few days using the systems already in place.
R – *Relevant*: Sustains all the project’s efforts — keeping data current, following up on promotions, tracking brand growth, and improving operations over time.
T – *Time-bound*: Onboard the intern within the next 4 weeks and have them fully trained and independently updating dashboards within 1 month.

Why These Recommendations Matter

These steps are practical and achievable — and more importantly, they're based directly on what the data showed. Starting with something simple like selling milk under the store's brand can create instant impact. Over time, if Jain Ki Dukan's own products become more visible and better aligned with what customers actually want, the store will not only make better margins but also build a stronger local identity and more loyal customers.

Looking ahead, continuing to track sales trends, test small changes, and revisit performance regularly will be key. The insights from this project show that data isn't just about what happened—it's a tool for shaping what comes next. With the right focus and small, informed moves, the store can keep improving operations, reducing waste, and making smarter choices that benefit both the business and its customers.

This project showed the real power of data in the business world, that it can be used to analyse the past and predict the future, which can be applied to foster business growth. With the right approach, data can reveal patterns that aren't obvious at first glance, highlight strengths worth doubling down on, and expose weak spots that need attention.

5. Appendix:

Google drive link:
https://drive.google.com/drive/folders/11fWhHCjA7C_1u9FGdMwgNtv7pPzNg1Lc?usp=sharing

Link to dataset:

https://drive.google.com/drive/folders/1y_TOjNF9im6cf19jIb0IkuxxHzCqsHl8?usp=sharing

Link to Analysis codes:

https://drive.google.com/drive/folders/1PquXMVdnXqfXNfgXMYj3mlX4Z_a-SjUp?usp=sharing